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CT pulmonary angiography-based scoring system to predict the prognosis of acute pulmonary embolism

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Abstract

Background: The purpose is to develop a comprehensive risk-scoring system based on CT findings for predicting 30-day mortality after acute pulmonary embolism (PE), and to compare it with PE Severity Index (PESI).

Materials and methods: The study included consecutive 1698 CT pulmonary angiograms (CTPA) positive for acute PE performed at a single institution (2003-2010). Two radiologists independently assessed each study regarding clinically relevant findings and then performed adjudication. These variables plus patient clinical information were included to build a LASSO logistic regression model to predict 30-day mortality. A point score for each significant variable was generated based on the final model. PESI score was calculated in 568 patients who visited the hospital after 2007.

Results: Inter-reader agreements of interpretations were >95% except for septal bowing (92%). The final prediction model showed superior ability over PESI (AUC = 0.822 vs 0.745) for predicting all-cause 30-day mortality (12.4%). The scoring system based on the significant variables (age (years), pleural effusion (+20), pericardial effusion (+20), lung/liver/bone lesions suggesting malignancy (+60), chronic interstitial lung disease (+20), enlarged lymph node in thorax (+20), and ascites (+40)) stratified patients into 4 severity categories, with mortality rates of 0.008% in class-I (≤ 50 pt), 3.8% in class-II (51-100 pt), 17.6% in class-III (101-150 pt), and 40.9% in class-IV (> 150 pt). The mortality rate in the CTPA-high risk category (class-IV) was higher than those in the PESI's high risk (27.4%) and very high risk (25.2%) categories.

Conclusion: The CTPA-based model was superior to PESI in predicting 30-day mortality. Incorporating the CTPA-based scoring system into image interpretation workflows may help physicians to select the most appropriate management approach for individual patients.

Keywords: Computed tomography; Mortality; Prediction; Pulmonary embolism; Risk assessment.

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